Anurag Pradhan

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Portfolio | Google Scholar

Professional Summary

Computer Science Researcher specializing in Deep Learning, Medical AI, and Generative Models. Summer Research Intern at IIT Mandi under Prof. Aditya Nigam, engineering MRI-to-synthetic CT reconstruction pipelines using Diffusion Models, GANs, and Mamba. Proficient in PyTorch, CNNs, Transformers, and VLM architectures. Co-authored an IJCNLP–AACL 2025 publication.

Education

Vellore Institute of Technology | *B.Tech. in Computer Science* **Aug 2023 – Jun 2027** | *Chennai, India*

Sri Venkateswara Jr. College | *Class 12th (MPC)* **Jul 2021 – Jun 2023** | *Visakhapatnam, India*

Experience

Indian Institute of Technology Mandi (IIT Mandi) **May 2026 – Present**
Research Intern | *Deep Learning, Diffusion Models, Mamba, GANs, Vision Encoders* *Mandi, Himachal Pradesh*
- Spearheading research on MRI-to-synthetic CT (sCT) reconstruction for radiotherapy planning under Prof. Aditya Nigam, implementing Diffusion Models, GANs, and Mamba architectures.
- Formulating a novel sliding-window DDPM inference technique for high-fidelity 3D volume synthesis, reducing full-volume MAE from ∼300 HU to ∼98 HU across 4 anatomical regions on CT scans.

Team Ignition, VIT Chennai **Apr 2024 – Apr 2026**
Flight Software Lead | *Arduino, C/C++, PID Control, Image Processing, Deep Learning* *On Site*
- Led flight software development across 3 CanSat competition entries, winning all 3.
- Devised fin actuator control, Kalman-filter-based state estimation, PID stability algorithms, and real-time onboard video stabilization, improving telemetry reception reliability by 15% on low-power hardware.

National Aluminium Company Ltd (NALCO) **Jun 2025 – Jul 2025**
Computer Science Intern | *Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation* *Damanjodi, Odisha*
- Deployed Respiguard AI, a YOLOv5-based lung disease prediction system for mine worker health screening.
- Fine-tuned on NIH ChestX-ray14 dataset (112,120 images, 14 classes); improved detection reliability via targeted augmentation, class-weighted loss, and iterative validation.

Projects

DrishtiAI: On-Device Spatial Vision Assistant for BLV | *Kotlin, SmolVLM2, llama.cpp, ONNX*
- Architected a 100% offline multimodal Android assistant using SmolVLM2-500M, llama.cpp (C++ JNI), and ONNX Runtime for real-time spatial vision and visual QA for blind and low-vision users.
- Fused Depth-Anything dense metric depth with YOLO-World v2 and CLIP zero-shot detection to compute 3D distances, outputting clock-face spatial guidance (e.g., *"door at 12 o'clock, 1.8m"*).
- Accelerated edge inference via ARMv8.4 I8MM SIMD, single-crop 384px prefill (≈256 tokens), and core pinning (13.5+ tok/s); added volume-key FSM, pocket protection, and tactile haptic alerts.

CMCA-ActionNet – Spatio-Temporal Multimodal Action Detection | *PyTorch, MediaPipe, Optical Flow*
- Engineered a multimodal human action recognition system achieving 84% accuracy on the JHMDB dataset.
- Formulated a Cross-Modal Causal Attention (CMCA) module fusing RGB, optical flow, and MediaPipe pose landmarks, raising action classification accuracy by 6.2% compared to single-modal baselines.
- Embedded bidirectional temporal attention and an uncertainty-aware prediction head for robust action recognition, cutting false positive rates by 15% on unconstrained video sequences.

Uncertainty-Aware Feedback System for MRI-to-sCT Synthesis | *Wavelet U-Net++, Deep Ensembles*
- Architected a dual-pass 2.5D Wavelet U-Net++ framework with 3D discrete wavelet transforms (db1) and deep ensembles to predict probabilistic mean (μ) and heteroscedastic aleatoric variance (σ^2).
- Decomposed variance into aleatoric and epistemic uncertainty; built a tissue-stratified isotonic regression calibrator, reducing Expected Normalized Calibration Error (ENCE) by 99.9% ($ENCE < 0.19$).
- Engineered a 7-channel residual refinement network with uncertainty-weighted edge loss, cutting Bone MAE by 83.4 HU, DVH D_{50} dose error by 56.8%, and raising 2%/2mm Gamma Pass Rate by 10.4% on 60 patients.

Conference Publications

Towards Blind and Low-Vision Accessibility of Lightweight VLMs and Custom LLM-EvalsPaper Link
- S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP–AACL 2025), Mumbai, India. Acceptance rate: ∼22%.

Technical Skills

Programming Languages: Python, C, C++, Java, Kotlin
Core Expertise & Research: VLMs, LLMs, EdgeAI, Generative AI, Action Recognition, Quantization, Medical AI
Frameworks & Libraries: PyTorch, TensorFlow, Transformers, llama.cpp, SimpleITK, PyWavelets
Tools & Platforms: Git, CUDA, ONNX, Hugging Face Hub